

Probability & Statistics (MT-2005)

Descriptive Statistics

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Distribution Shapes/ Measure of Asymmetry

Skewness

Definition: Measures the asymmetry of a data distribution around the mean. **Formula:**

$$\text{Skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{\frac{3}{2}}}$$

Alternate Formula:

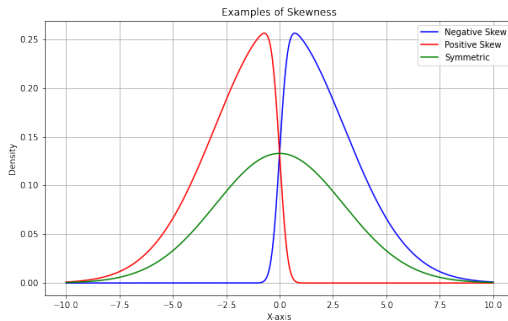
$$\text{Skewness} = \frac{\sum_{i=1}^n (x_i - \bar{x})^3}{n \cdot \sigma^3}$$

Interpretation:

- **Symmetric:** Skewness = 0
- **Positively Skewed:** Mean > Median
- **Negatively Skewed:** Mean < Median

Example: Dataset {1, 2, 3, 10}: Positively skewed.

Skewness Visualization



Kurtosis

Definition: Measures the "tailedness" or peakedness of a data distribution.

Formula:

$$\text{Kurtosis} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\right)^2}$$

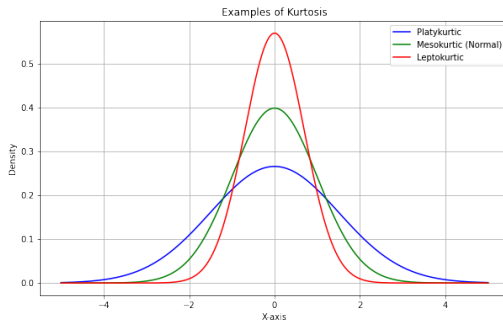
Alternate Formula:

$$\text{Kurtosis} = \frac{\sum_{i=1}^n (x_i - \bar{x})^4}{n \cdot \sigma^4}$$

Types:

- **Mesokurtic:** Normal distribution (Similar to normal curve) (Kurtosis = 3).
- **Leptokurtic:** Heavy tails, sharp peak (Taller and narrower) (Kurtosis > 3).
- **Platykurtic:** Light tails, flat peak (Flatter and broader) (Kurtosis < 3).

Kurtosis Visualization



More about Shapes of Distributions

J-shaped, L-shaped, and U-shaped Distributions

The terms "J-shaped," "L-shaped," and "U-shaped" distributions refer to the general shapes of the probability density functions (or histograms) of data distributions. These terms help describe the overall structure and skewness of the data.

J-shaped Distribution

A J-shaped distribution is one where the data starts off with a relatively low frequency in the lower values, increases sharply, and then flattens out at higher values, forming a shape resembling the letter “J.” It has:

- A low frequency or low values at the start of the distribution,
- A sharp rise at a certain point,
- A tail or a slower increase (sometimes flat) as the distribution moves toward larger values.

Example: A population that starts from a small base and rapidly increases over time, such as a new technology adoption or early-stage market expansion.

L-shaped Distribution

An L-shaped distribution refers to a distribution that starts high (often at the beginning of the data range) and then drops sharply before leveling off, forming a shape that looks like the letter “L.” It has:

- A high frequency at the lower end of the distribution,
- A sharp drop-off in frequency as you move toward higher values,
- A flat tail on the right side where the frequency becomes relatively constant.

Example: A large number of customers make initial purchases, but very few make repeat purchases (e.g., in some types of product life cycle models).

U-shaped Distribution

A U-shaped distribution resembles the shape of the letter “U,” where:

- The distribution has a higher frequency at both ends, with a lower frequency in the middle,
- It typically reflects a situation where the values in the extreme categories (both lower and higher) are more likely than those in the middle range.

Example: In some surveys, people with extreme views (either strongly agreeing or strongly disagreeing) are more common than those with moderate or neutral opinions (e.g., extreme political views).

Measures of Position

Measures of Position

- Measures of position indicate the location of a value relative to the rest of the data.
- Common measures of position:
 - Percentiles
 - Quartiles
 - Deciles
 - Standard Scores (Z-Scores)
- Useful for comparing individual values to a dataset.

Quartiles for Ungrouped Data

- Quartiles divide data into four equal parts.
- Key Quartiles:
 - Q_1 : First quartile (25th percentile)
 - Q_2 : Second quartile (Median or 50th percentile)
 - Q_3 : Third quartile (75th percentile)

Formula for Quartile Position:

$$Q_k = \left(\frac{k(n+1)}{4} \right)^{th} \text{ value, } k = 1, 2, 3 \quad (1)$$

where n is the number of observations.

Example: Given a dataset: {12, 5, 8, 21, 14}

Step 1: Arrange the Data

- Sort the data in ascending order: {5, 8, 12, 14, 21}

Quartiles for Ungrouped Data Cont'd...

Step 2: Identify Quartile Positions

- First Quartile (Q_1): $Q_1 = \frac{n+1}{4}$ -th value
- Third Quartile (Q_3): $Q_3 = \frac{3(n+1)}{4}$ -th value where n = Number of data points.

Step 3: Locate Quartiles

- Interpolate if the position is not an integer.
- Example with $n = 5$:

$$Q_1 = \frac{(5+1)}{4} = 1.5 \quad Q_3 = \frac{3(5+1)}{4} = 4.5$$

Why Use $n + 1$ Instead of n ?

Reason:

- Using $n + 1$ accounts for positions between ranks more evenly, especially for smaller datasets.

Example: Dataset {4, 7, 10, 15, 20}

- **Using $n = 5$:**

$$Q_1 = \frac{5}{4} = 1.25 \quad \text{Interpolated Value: 4.75}$$

$$Q_3 = \frac{3(5)}{4} = 3.75 \quad \text{Interpolated Value: 13.75}$$

- **Using $n + 1 = 6$:**

$$Q_1 = \frac{6}{4} = 1.5 \quad \text{Interpolated Value: 5.5}$$

$$Q_3 = \frac{3(6)}{4} = 4.5 \quad \text{Interpolated Value: 17.5}$$

Conclusion:

- $n + 1$ provides a more accurate representation of quartiles.

Quartiles for Grouped Data

Formula:

$$Q_k = L + \left(\frac{\left(\frac{k \cdot N}{4} \right) - CF}{f} \right) \cdot w$$

Where:

- L : Lower boundary of the class containing P_k
- N : Total frequency
- CF : Cumulative frequency before the class
- f : Frequency of the class containing P_k
- w : Class width
- $k = 1, 2, 3$ for Q_1, Q_2, Q_3 .

Deciles for Ungrouped Data

- Deciles divide the data into 10 equal parts.
- The k -th decile (D_k) represents the value below which $k\%$ of the data falls, where $k = 1, 2, \dots, 9$.

Formula:

$$D_k = \left(\frac{k(n+1)}{10} \right)^{th} \text{ value, } k = 1, 2, \dots, 9 \quad (2)$$

Deciles for Grouped Data

Formula:

$$D_k = L + \left(\frac{\left(\frac{k}{10} \cdot N \right) - CF}{f} \right) \cdot w$$

Where:

- L : Lower boundary of the class containing D_k
- N : Total frequency
- CF : Cumulative frequency before the class
- f : Frequency of the class containing D_k
- w : Class width
- $k = 1, 2, \dots, 9$

Percentiles for Ungrouped Data

- Percentiles divide data into 100 equal parts.
- The k -th percentile (P_k) represents the value below which $k\%$ of the data falls.

Formula:

$$P_k = \left(\frac{k(n+1)}{100} \right)^{th} \text{ value, } k = 1, 2, \dots, 99 \quad (3)$$

Percentiles for Grouped Data

Formula:

$$P_k = L + \left(\frac{\left(\frac{k}{100} \cdot N \right) - CF}{f} \right) \cdot w$$

Where:

- L : Lower boundary of the class containing P_k
- N : Total frequency
- CF : Cumulative frequency before the class
- f : Frequency of the class containing P_k
- w : Class width
- $k = 1, 2, \dots, 99$

Percentile Rank: Definition and Formula

Definition: The percentile rank (PR) of a score X is the percentage of scores in a distribution that are less than X , plus half the proportion of scores equal to X .

Formula:

$$\text{PR} = \frac{\text{Number of scores below } X + 0.5 \cdot \text{Frequency of } X}{\text{Total number of scores}} \times 100$$

Example Dataset

Dataset: [45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]

Find the percentile rank of $X = 75$.

- Total number of scores (n): 11
- Scores below 75: 6
- Frequency of 75: 1 (appears once in the dataset)

Calculation of Percentile Rank

Using the formula:

$$PR = \frac{\text{Number of scores below } X + 0.5 \cdot \text{Frequency of } X}{\text{Total number of scores}} \times 100$$

Substituting the values:

$$PR = \frac{6 + 0.5 \cdot 1}{11} \times 100$$

Simplify:

$$PR = \frac{6.5}{11} \times 100 \approx 59.09$$

Percentile Rank: 75 is at approximately the 59th percentile.

Interpretation of the Result

- 59% of the scores in the dataset are less than 75.
- 75 is positioned slightly above the middle of the dataset.

Why add $0.5 \cdot \text{Frequency of } X$?

- Ensures a fair inclusion of X in its frequency group.
- Accounts for X being part of the dataset, without overstating its contribution.

Standard Scores (Z-Scores)

- A Z-Score indicates how many standard deviations a data point is from the mean.

Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Where:

- x : Value in the dataset
- μ : Mean of the dataset
- σ : Standard deviation of the dataset

Interpretation:

- $Z > 0$: Value is above the mean.
- $Z < 0$: Value is below the mean.
- $Z = 0$: Value is equal to the mean.

Semi Interquartile Range/Quartile Deviation

Definition: Quartile Deviation measures the spread of the middle 50% of the data, focusing on the interquartile range (IQR). It is an *absolute measure of dispersion*.

Formula:

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

where:

- Q_1 : First Quartile (25% of the data below this value)
- Q_3 : Third Quartile (75% of the data below this value)

Characteristics:

- Represents half the spread of the middle 50% of data.
- Not affected by extreme values (robust to outliers).
- Provides insight into the central dispersion of the dataset.

Definition of the Midhinge

- The **Midhinge** is a measure of central tendency that is the average of the first and third quartiles (Q_1 and Q_3).
- Formula:

$$\text{Midhinge} = \frac{Q_1 + Q_3}{2}$$

- It provides a sense of the center of the middle 50% of the data.

Key Features of the Midhinge

- The Midhinge is sensitive to the spread of the data since it depends on the quartiles.
- It is less affected by extreme values (outliers) compared to the mean.
- Often used in exploratory data analysis as a complement to other measures like the median.

Relation Between Midhinge and Median

- **Symmetric Distributions:**

- The Midhinge and Median are equal.

- **Skewed Distributions:**

- In a left-skewed distribution, the Midhinge is greater than the Median.
- In a right-skewed distribution, the Midhinge is less than the Median.
- The difference between the Midhinge and the Median can indicate the skewness of the data.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)

Definition: EDA involves preliminary checks to understand the data's nature, identify anomalies, and select appropriate statistical methods.

Key Tools:

- **Stem-and-Leaf Plot:** Visualizes data distribution in a compact form.
- **Box Plot:** Highlights data spread, median, and potential outliers.

Historical Note: EDA was introduced by John Tukey in 1977.

Stem-and-Leaf Plot

A **stem-and-leaf plot** is a method of organizing data in a way that shows the distribution of values while retaining the original data. It is a graphical representation that separates each data point into two parts: the stem and the leaf. The stem typically represents the leading digits (such as tens, hundreds), and the leaf represents the trailing digits (such as ones, tenths).

Example # 7: The following is the stem-and-leaf plot for the dataset: {65, 67, 69, 72, 73, 76, 77, 78, 81, 84, 85, 88, 92, 95}.

Stem	Leaves
6	5, 7, 9
7	2, 3, 6, 7, 8
8	1, 4, 5, 8
9	2, 5

Back-to-Back Stem-and-Leaf Plot

Example # 8:

Dataset 1: {67,72,78,65,88,85,82,95,92,76,84,73,77,81,69}

Dataset 2: {66,70,74,80,83,87,89,91,93,96,75,78,82,86,90}

We compare two datasets using a back-to-back stem-and-leaf plot:

Leaves (Dataset 2)	Stem	Leaves (Dataset 1)
6, 6	6	5, 7, 9
0, 4, 5, 7	7	2, 3, 6, 7, 8
0, 2, 6	8	1, 2, 4, 5, 8
0, 1, 3, 6	9	2, 5

Legend:

- **Stem:** Represents the tens digit.
- **Leaves:** Represents the units digit.

Box and Whisker Plot

A box and whisker plot is a graphical representation that displays the distribution of a data set through five summary statistics: the minimum, first quartile (Q_1), median, third quartile (Q_3), and maximum.

- **Minimum:** The smallest data point within a specified range (excluding outliers).
- **First Quartile (Q_1):** The 25th percentile; 25% of the data points lie below this value.
- **Median:** The middle value of the data set, dividing the data into two equal halves.
- **Third Quartile (Q_3):** The 75th percentile; 75% of the data points lie below this value.
- **Maximum:** The largest data point within the specified range (excluding outliers).
- **Whiskers:** Lines that extend from the box to the smallest and largest values within 1.5 times the interquartile range (IQR).
- **Outliers:** Data points that fall outside of the whiskers' range ($1.5 * \text{IQR}$).

Identification of Outliers in Box and Whisker Plot

Outliers are identified based on their position relative to the whiskers:

- **Step 1: Calculate the IQR.**

$$\text{IQR} = Q_3 - Q_1$$

- **Step 2: Determine the lower and upper bounds.**

$$\text{Lower Bound/Fence} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound/Fence} = Q_3 + 1.5 \times \text{IQR}$$

- **Step 3: Identify outliers.** Any data points below the lower fence or above the upper fence are considered outliers.

Outliers are plotted as individual points outside the whiskers in the box and whisker plot.

Box and Whisker Plot: Example

The given dataset is:

12, 7, 3, 18, 21, 13, 8, 17, 5, 10, 15, 19, 6, 9, 11

Step 1: Organize Data Arrange the data in ascending order:

3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 17, 18, 19, 21

Step 2: Calculate the Five-Number Summary

- **Minimum (Min):** The smallest value in the dataset is 3.
- **Maximum (Max):** The largest value in the dataset is 21.
- **Median:** The middle value is 10 (as there are 15 data points).
- **Q_1 (First Quartile):** The median of the lower half (excluding the median value) is 6.
- **Q_3 (Third Quartile):** The median of the upper half (excluding the median value) is 17.

Step 3: Interquartile Range (IQR)

The Interquartile Range (IQR) is calculated as:

$$IQR = Q_3 - Q_1 = 17 - 6 = 11$$

Step 4: Identify Outliers

Outliers are data points that lie outside the range:

$$\text{Lower Fence} = Q_1 - 1.5 \times IQR = 6 - 1.5 \times 11 = 6 - 16.5 = -10.5$$

$$\text{Upper Fence} = Q_3 + 1.5 \times IQR = 17 + 1.5 \times 11 = 17 + 16.5 = 33.5$$

Since no data points fall outside the range $[-10.5, 33.5]$, there are no outliers in this dataset.

Summary of the Box and Whisker Plot Components

- **Minimum (Min):** 3
- **Q_1 (First Quartile):** 6
- **Median:** 10
- **Q_3 (Third Quartile):** 17
- **Maximum (Max):** 21
- **Outliers:** None

Python Code: Box and Whisker Plot Example

Here is the Python code for creating a Box and Whisker Plot:

```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Sample data: You can replace this with your own dataset
data = [12, 7, 3, 18, 21, 13, 8, 17, 5, 10, 15, 19, 6, 9,
        11]

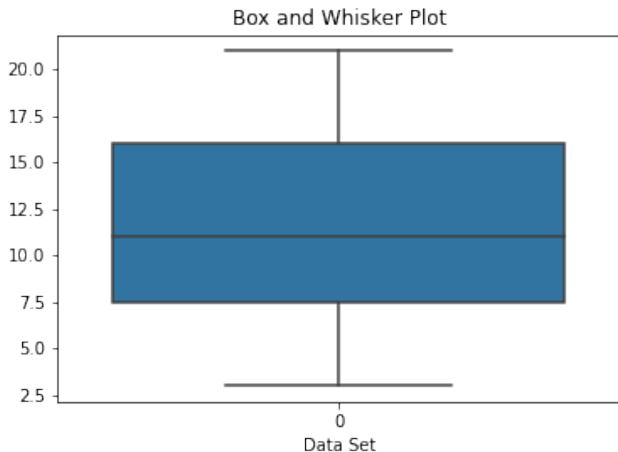
# Create a box plot using Seaborn
sns.boxplot(data=data)

# Add title and labels
plt.title('Box and Whisker Plot')
plt.xlabel('Data Set')

# Save the plot
plt.savefig('Box n Whisker.png')
plt.close()
```


Box and Whisker Plot Visualization

After running the code, you will get the following Box and Whisker Plot:



Example: Box and Whisker Plot with Outliers

Consider the following data set with potential outliers:

Data: 12, 7, 3, 18, 21, 13, 8, 17, 5, 10, 15, 19, 6, 9, 11, 50

First, calculate the quartiles and IQR:

- **Q_1 (First Quartile):** 6
- **Median (Q_2):** 12
- **Q_3 (Third Quartile):** 17
- **IQR:** $17 - 6 = 11$
- **Lower Bound:** $6 - 1.5 \times 11 = -9.5$
- **Upper Bound:** $17 + 1.5 \times 11 = 32.5$

Example of Box and Whisker Plot with Outliers cont'd...

Since the value 50 is greater than the upper bound (32.5), it is considered an outlier.

The box plot will show:

- The box between Q_1 (6) and Q_3 (17), with a median line at 12.
- The whiskers extending from Q_1 (6) to the maximum value within the upper bound (21).
- The value 50 plotted as an individual point beyond the upper whisker, marked as an outlier.

Box Plot Fences

- **Lower Inner Fence:**

$$Q_1 - 1.5 \times \text{IQR}$$

- **Upper Inner Fence:**

$$Q_3 + 1.5 \times \text{IQR}$$

- **Lower Outer Fence:**

$$Q_1 - 3 \times \text{IQR}$$

- **Upper Outer Fence:**

$$Q_3 + 3 \times \text{IQR}$$

- **Interpretation:**

- Points between inner and outer fences are **mild outliers**.
- Points beyond outer fences are **extreme outliers**.

Python for Descriptive Statistics

Healthcare Dataset Overview

Dataset: Patient health metrics including Age, BMI, Blood Pressure, and Heart Rate.

Patient ID	Age (Years)	BMI	BP (mmHg)	Heart Rate (bpm)
101	25	22.4	120	72
102	34	27.8	138	80
103	52	30.5	145	76
104	45	24.7	130	88
105	30	29.3	140	85

Table: Healthcare Data

Summary:

- **Count:** 5 patients.
- **Mean:**
 - Age: 37.2 years
 - BMI: 26.94
 - Blood Pressure: 134.6 mmHg
 - Heart Rate: 80.2 bpm
- **Standard Deviation:**
 - Age: 10.82
 - BMI: 3.55
 - Blood Pressure: 10.72
 - Heart Rate: 6.57

Python Code for Healthcare Data Analysis

```
import pandas as pd

# Sample healthcare data
data = {
    "Patient_ID": [101, 102, 103, 104, 105],
    "Age": [25, 34, 52, 45, 30], # in years
    "BMI": [22.4, 27.8, 30.5, 24.7, 29.3], # Body Mass
        Index
    "Blood_Pressure_Systolic": [120, 138, 145, 130, 140], #
        mmHg
    "Heart_Rate": [72, 80, 76, 88, 85] # beats per minute
}

# Create a DataFrame
df = pd.DataFrame(data)

# Display the dataset
print("Healthcare Dataset:\n", df)
```


Python Code for Healthcare Data Analysis cont'd...

```
# Generate descriptive statistics for the data
print("\nDescriptive Statistics:\n", df.describe())

# Calculate additional statistics
print("\nMedian:\n", df.median(numeric_only=True))
print("\nMode:\n", df.mode(numeric_only=True))
print("\nVariance:\n", df.var(numeric_only=True))
print("\nStandard Deviation:\n", df.std(numeric_only=True))
```

Python Code for Healthcare Data Analysis cont'd...

```
# Column-wise summary
print("\nColumn-Wise Summary:")
for column in df.columns[1:]: # Exclude Patient_ID
    print(f"\n{column} Summary:")
    print(f"    Min: {df[column].min()}")
    print(f"    Max: {df[column].max()}")
    print(f"    Range: {df[column].max() - df[column].min()}")
    print(f"    Mean: {df[column].mean()}")
    print(f"    Median: {df[column].median()}")
    print(f"    Standard Deviation: {df[column].std()}")
```

Calculated Statistics:

- **Median:**

- Age: 34 years
- BMI: 27.8
- Blood Pressure: 138 mmHg
- Heart Rate: 80 bpm

- **Mode:**

- All columns: No repetition (unique values).

- **Variance:**

- Age: 117.2
- BMI: 12.6
- Blood Pressure: 114.8
- Heart Rate: 43.2

Insights:

- **Age:**

- Min: 25, Max: 52, Range: 27

- **BMI:**

- Min: 22.4, Max: 30.5, Range: 8.1

- **Blood Pressure:**

- Min: 120, Max: 145, Range: 25

- **Heart Rate:**

- Min: 72, Max: 88, Range: 16

Output from Python Code

Python Console Output:

Healthcare Dataset:

	Patient_ID	Age	BMI	Blood_Pressure_Systolic	Heart_Rate
0	101	25	22.4	120	72
1	102	34	27.8	138	80
2	103	52	30.5	145	76
3	104	45	24.7	130	88
4	105	30	29.3	140	85

Descriptive Statistics:

	Age	BMI	Blood_Pressure	Heart_Rate
Count	5.00	5.0	5.00000	5.000
Mean	37.20	26.94	134.60000	80.200
Std Dev	10.82	3.55	10.72211	6.574